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Demography past papers

health records and information technology (Kenya Medical Training College)

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KENYA MEDICAL TRAINING COLLEGE
FACULTY OF INFORMATION TECHNOLOGY AND MEDICAL EDUCATION
DEPARTMENT OF HEALTH RECORDS AND INFORMATION

YEAR ONE END OF SEMESTER ONE EXAMINATION FOR SEPTEMBER 2021 (UPGRADING)

DIPLOMA IN HEALTH RECORDS AND INFORMATION TECHNOLOGY

PAPER: MEDICAL DEMOGRAPHY I

DATE: 4th February 2022 **TIME: 3 HOURS (9:00AM – 12:00 PM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (20 Multiple Choice Questions)
   - Section 2 (6 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write your **STUDENT NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

STUDENT NUMBER .....

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SECTION ONE: MULTIPLE CHOICE QUESTIONS (20 MARKS)

1. The measures put in place by governments to control the number of births in a country are known as……

A. Morbidity C. Fertility
B. Population Policy D. Family control

2. The following are examples of pull factors apart from one, which is it?

A. Good infrastructure C. Availability of food
B. Good climate D. Inadequate rainfall

3. Which of the following is not a feature of civil registration systems in Kenya?

A. It is continuous
B. It is compulsory
C. Targets only specific population
D. It is legal.

4. The increase or decrease in the number of people in a defined territory is defined as……

A. Population size
B. Population change
C. Population structure
D. Population distribution

5. Which of the following is not a source of demographic data?

A. Sample surveys C. Journals
B. Census reports D. hospital records

6. The proportion of males to females in a population is defined as……

A. Sex ratio C. sex structure
B. Sex distribution D. sex composition → cohort group of ppl sharing same characteristic eg age

7. Which of the following does not have effect on the general sex ratio……

A. Sex ratio at birth
B. Migration patterns among the sexes
C. Fertility rates
D. Sex mortality at birth

8. Which of the following is not an element of the population balancing equation?

A. Population at the start
B. Natural increase
C. Midyear population
D. Net migration

9. Demographic transition theory took place in…….. stages

A. 2 C.5
B. 3 D.4

10. The ratio of economically inactive individuals to economically active individuals in a community is termed as……..

A. Sex ratio C.age dependency ratio
B. Doubling time D. age ratio

11. The number of years a population takes to be twice its size from the current is termed as……..

A. Doubling time
B. Growth period
C. desiderata
D. natural increase

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12. Which of the following is not an effect of population growth?
A. Requires more investment in social amenities
B. Food shortage
C. Lowers standard of living
D. Increases food production
13. Which of the following ages does not correspond with the right set?
A. Below 1 year-infancy
B. Childhood-1-14 years
C. Adulthood-35 years and above
D. Neonate-above 1 year
14. In a population pyramid the males are shown on the……side
A. Middle
B. Left
C. Right
D. Both left and right
15. Which of the following does not affect the population of a place……
A. Migration out
B. Number of births
C. Number of deaths
D. Stability
16. The type of census that counts people with legal right to belong to a place is known as……..
A. De facto C.de jure
B. Surveys Civil registration
17. Which of the following is not a feature of census?
A. sponsorship
B. defined territory
C. sponsorship
D. Counts groups
18. The following are sources of content error except one, which is it……..
A. Respondents giving wrong information
B. Wrong classification by enumerator
C. Time of collection
D. Editing process
19. The form of age misreporting where people report ages ending with 0 or 5 is known as…….
A. Age heaping
B. Age shifting
C. Whipples index
D. Age adjustment
20. The frequency of marriages in a population is termed as……
A. Nuptiality C. Fertility
B. Fecundity D. Mortality

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SECTION TWO: SHORT ANSWER QUESTIONS (30 MARKS)

Pop change = Natural increase + Net migration

$P_t = P_{t_0} + (B - D) +/ IN - out$ migration

Pop change, Previous pop

1. Describe the basic demographic equation.(5 marks)
2. Describe the Kenya's population policy (5 marks)
3. List the 5 sources of census data (5marks)
4. List any 5 features of civil registration systems (5marks)
5. State the effects of population growth (5marks) → large pop create unemployment. Decline of trend of agricultural drip. rapid pop growth decline social infrastructure. rapid pop grow creat food shortage. growth pop lower living standard
6. List any 5 methods of population control (5marks) one child legislation. promote ffp. Distract early mamage. woman empowerment. Consider adoption art policy

SECTION THREE: LONG ANSWER QUESTIONS (20 MARKS)

1. The population of Kenya was 28 million in 2012 with a growth rate of 3% p.a, the female population was 30 % of the total population (10 marks)
   a. What was the sex ratio
   b. When will the population double
   c. What was the doubling time
   d. List any 5 factors that could have made the population to grow faster.
2. Describe any 5 methods of population control (10 marks)

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MURANG'A MEDICAL TRAINING COLLEGE

DEPARTMENT OF HEALTH RECORDS AND INFORMATION TECHNOLOGY

September 2010 DIPLOMA CLASS (PARALLEL)

SEMESTER 4 MEDICAL DEMOGRAPHY EXAMINATION 22nd /03/2011

Important information

1. Read carefully and answer all the questions clearly.
2. Provide all the answers and workings in the answer sheet provided and number all questions accordingly. **Note:** any un numbered question will not me marked.
3. Write the student number on every answer sheet used.
4. Time given is three (2) hours

Question 1

Briefly define the following terms as used in demography (12marks)

i) Infant mortality rate

ii) Neonatal mortality rate

iii) Child mortality rate

iv) Post neonatal mortality rate

v) Maternal mortality rate

vi) General fertility rate

Question 2

The following data relates to a number of women and births in Kenya in 1973. Midyear population was reported to be 96, 398,000 (18 marks)

Age group	No. of women in thousands	No. of births
15-19	3777000	756,900
20-24	3101000	1,046,000
25-29	2636000	819,500
30-34	2161000	565,100
35-39	1773000	353,200
40-44	1484000	141,600
45-49	1222000	16,500
	16,154,000	3,698,800

a) Calculate crude birth rate in Kenya in 19~~00~~ 73

b) General fertility rate

c) Total fertility rate

d) Interpret your answer in c

e) Given that the number of children aged 0-4 years is 8,730,000 calculate child woman ratio

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Question 3

Using direct method of standardization, standardize both city A and B using the two populations as your standard. Express your answer in percentage. (10 marks)

	City A		City B	
Age group	Population size	no of deaths	Population size	no of deaths
Children	1,000	1	4,000	8
Youth	5,000	50	5,000	100
Old	4,000	400	1,000	200

Question 4

Applying indirect method of standardization, standardize the mortality rate in country B using country A as the standard population and interpret your answer.
NB; Death rates should be per 100,000 (10 marks)

POPULATION IN COUNTRY A AND COUNTRY B (FEMALES)

	Country A		Country B	
AGE GROUP	FEMALES	NO. OF DEATHS	FEMALES	NO OF OF DEATHS
0-4	1,900,000	1,406	26,000	21'
5-14	3,100,000	186	30,000	2
15-44	9,400,000	1,786	127,000	27
45-64	4,900,000	7,350	25,000	42
65+	2,000,000	17,400	5,000	48

Question 5

a) Discuss five (5) importance of a life table in Public health (10 marks)

b) Define the meaning of the following functions of a life table (10 marks)

i) ${}_nd_x$

ii) ${}_nq_x$

iii) ${}_np_x$

iv) l_x

vi) t_x

vii) d_x

viii) ${}_nm_x$

ix) ${}_na_x$

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KENYA MEDICAL TRAINING COLLEGE
DEPARTMENT OF HEALTH RECORDS AND INFORMATION
DIPLOMA IN HEALTH RECORDS AND INFORMATION TECHNOLOGY
FINAL QUALIFYING EXAMINATION
SUBJECT: DEMOGRAPHY
TIME ALLOWED: 3 HOURS

DATE: 2/07/2014

INSTRUCTIONS

1. Write your Examination number at the right hand corner of the answer booklet
2. Be clear. precise and orderly in all your answers
3. Answer all Questions
4. Systematically show your computations
5. CHEATING in the examination will lead to disqualification from the entire examination.
6. MOBILE PHONES are not allowed in the examination room.

SECTION ONE

Indicate if the following statements are TRUE or FALSE in the answer booklet (10 Marks)

i. Birth certificate issuing is not proof of demographic data collection
ii. National Identification cards do not constitute documentation of Demographic data
iii. Family planning services are strategic events that determine fertility rate of a community
iv. Infant mortality rate have direct indication of population growth in a community
v. Leaving family size determination in the hands of parents is a better population control policy
vi. The basic demographic Equation does not consider time when being expressed
vii. Demographers use birthdays celebrations as an individual's specific age
viii. Grouping of ages in Demography may be chosen according to the purpose of observation on the data collection
ix. A population may be stated as intermediate if the Median falls between 20 and 29 years
x. In Demographic studies life expectancy is the sum of events in one's life

SECTION TWO- Short answers

Indicate the answer for blank spaces provided in the answer booklet (10 Marks)

i. The skill used to validate challenges of data from two geographical areas and enable comparison ---
ii. The most commonly used five years interval of age grouping in Demography is referred to as ----------
iii. The section of the population that is sixty five (65)years and above is referred to as ---------- Proportion
iv. In demographic terms a Population characterized by those aged less than 15 years being 40% or more is referred to as ---------- Population
v. In Demographic terms the type of population count that shows population actually present at the moment of census is referred to as ----------

SECTION THREE- Short Answer Questions

1. State three (3) most basic tools of measuring formal Demography (3 Marks)
2. State the three (3) major factors that influence population size change in most circumstances (3 Marks)
3. State any four (4)specific uses of demographic data (4 Marks)

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6. Homabay County 2009 census report extract showed population as extracted below:

Age (yrs)	Female	Males	Age	Females	Males
Under 1	4649	4283	40	5443	6321
1	4531	4130	41	535	419
2	7549	7132	42	851	670
3	7150	6705	43	520	359
4	7477	7069	44	504	321
5	6267	6006	45	1917	1788
6	7390	6719	46	578	376
7	6012	5501	47	528	301
8	7086	6422	48	884	647
9	4871	4570	49	507	291
10	8510	7566	50	3884	4497
11	4713	4219	51	346	260
12	6484	5285	52	560	387
13	4887	4321	53	479	234
14	5057	4129	54	481	223
15	6088	5041	55	956	821
16	4593	4239	56	511	318
17	3580	3248	57	346	185
18	5321	5035	58	356	226
19	2860	2762	59	201	115
20	7569	7730	60	2409	2596
21	1869	1481	61	167	154
22	2938	2496	62	281	148
23	2159	1827	63	276	141
24	2385	1807	64	226	89
25	5005	5638	65	488	424
26	2286	2038	66	235	87
27	2225	1996	67	222	127
28	2627	2891	68	146	107
29	1630	1376	69	88	63
30	7143	8202	70	1297	1500
31	989	651	71	110	65
32	1507	1215	72	137	90
33	1114	856	73	108	57
34	876	659	74	107	47
35	2810	3255	75	246	196
36	1093	905	76	136	56
37	945	764	77	87	37
38	1252	1198	78	114	71
39	761	555	79	38	47

Children <1

Elderly 6

sex ratio = M

Age dependen

a) Using this population extract, compute the following for Homabay County:

i. Sex Ratio PM/PF X100 (7 Marks)

ii. Age Dependence Ratio ~~Pop~~ 15yrs and below ~~above~~ 65+ / pop. 15-64 yrs X100 (7 Marks)

iii. Wipple's Index (8 Marks)

b) Give the meaning of the computed measurements in respect to Homabay County (3 Marks)

Ages 25, 30 —— 60 / bwn 23 to 62yrs X100

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Question 1

a. Crude birth rate is said to be a useful measure of fertility. Give at least 3 reasons as why this statement is true. (6 marks)

b. Explain how demographers take care of unknown ages or not stated ages in demographic d ta (5)

c. Ifcountries were to suspend the carrying out ofnational censuses, what data problems would t his present? ([4)

Question 2 (a)

Age group	England and Wales		Mauritius	
	Total population	Total malaria deaths.	Total population	Total malaria deaths
Under 1	200,000	120	100,000	287
1- 4	900,000	160	500,000	222
5-14	1,600,000	187	2,600,000	250
15-24	1,800,000	211	2,500,000	188
25-34	1,200,000	199	800,000	300
35-44	1,500,000	112	900,000	198

Age group	Kenya	
	Total population	Total malaria deaths.
Under 1	100,000	80
1- 4	950,000	124
5-14	1,400,000	148
15-24	1,600,000	198
25-34	1,800,000	175
35-44	1,400,000	96

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NOTES.

Use population for England and Wales as your standard population and the population for Mauritius and Kenya as your index population. Use 100,000 as your rate.

i) Standardize the malaria deaths for Mauritius. (3 marks)
ii) Calculate the crude malaria death rate for Mauritius.(3 marks)
iii) Standardize the malaria deaths for Kenya.(3 marks)
iv) Calculate the crude malaria death rate for Kenya (3 marks)
v) Comment on the standardized crude malaria death rates for these two countries. (6 marks)

b) Explain why age and any other variable are considered to be important in mortality analysis (7 marks)

$$C.D.R = \frac{\text{Deaths}}{\text{Total Pop'n.}} \times 100{,}000$$

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KMTC/QP-08/T1

KENYA MEDICAL TRAINING COLLEGE
DEPARTMENT OF HEALTH RECORDS AND INFORMATION
DIPLOMA IN HEALTH RECORDS AND INFORMATION
FINAL QUALIFYING SUPPLEMENTARY EXAMINATION
SUBJECT: DEMOGRAPHY DATE: 23/01/2014
TIME ALLOWED: 3 HOURS

INSTRUCTIONS

1. Write your examination number at the right hand corner of the answer book
2. Be clear, precise and orderly in your answers
3. Answer all Questions
4. CHEATING in the examination will lead to disqualification from the entire exam.
5. MOBILE PHONES are not allowed in the examination room.

1. State whether the following Demographic statements are TRUE or FALSE (10 Marks)
   i. When determining the median age of a given community the total population of a given age is not required.
   ii. Deaths occurring in a given community have no indication in the population growth rate of that community
   iii. Marital status is one of the details extracted during the delivery process for computation of fertility indicators in a community.
   iv. Determining population theory becomes an essential element of development of a given country
   v. Infant mortality rates are used to determine the health status of any community.
   vi. Mortality rate measures can only be expressed as proportion per a thousand population
   vii. Total fertility rate is the best indicator of fertility measure of any given community.
   viii. It is recommended that nations of the world conduct their population census every ten years.
   ix. Age heaping and age shifting are similar distortion practices in population studies
   x. Standardization techniques overcomes errors that were incurred during data collection

SECTION TWO

2. State the Five characteristics of determinants population studies (10 marks)
3. a) Describe the demographic transition theory (10 marks)
   b) Explain the following functions as used in life tables (10 marks)
   i) $_na_x$ vi) t_x – living above age x
   ii) $_nq_x$ – vii) d_x – Dying bt age x and x+1
   iii) $_np_x$ viii) $_nm_x$
   iv) l_x ix) $_nd_x$
   v) e_x x) $_nl_x$ – Living btwn age x and x+1
4. List the five qualities of Wipples index computation and their respective scales (10 Marks)

Quality	Index scales
i. High	Below 105
ii. Fairly	105 – 110
iii. Approximate	110 – 125
iv. Rough	125 – 175
v. very	above 175

(10Mark)

High accurate under 105
Fairly " 105 – 110
Approximate 110 – 125
Rough 125 – 175

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